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DNNGP, a deep neural network-based method for genomic prediction using multi-omics data in plants

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ABSTRACT

Genomic prediction is an effective way to accelerate the rate of agronomic trait improvement in plants. Traditional methods typically use linear regression models with clear assumptions; such methods are unable to capture the complex relationships between genotypes and phenotypes. Non-linear models (e.g., deep neural networks) have been proposed as a superior alternative to linear models because they can capture complex non-additive effects. Here we introduce a deep learning (DL) method, deep neural network genomic prediction (DNNGP), for integration of multi-omics data in plants. We trained DNNGP on four datasets and compared its performance with methods built with five classic models: genomic best linear unbiased prediction (GBLUP); two methods based on a machine learning (ML) framework, light gradient boosting machine (LightGBM) and support vector regression (SVR); and two methods based on a DL framework, deep learning genomic selection (DeepGS) and deep learning genome-wide association study (DLGWAS). DNNGP is novel in five ways. First, it can be applied to a variety of omics data to predict phenotypes. Second, the multilayered hierarchical structure of DNNGP dynamically learns features from raw data, avoiding overfitting and improving the convergence rate using a batch normalization layer and early stopping and rectified linear activation (rectified linear unit) functions. Third, when small datasets were used, DNNGP produced results that are competitive with results from the other five methods, showing greater prediction accuracy than the other methods when large-scale breeding data were used. Fourth, the computation time required by DNNGP was comparable with that of commonly used methods, up to 10 times faster than DeepGS. Fifth, hyperparameters can easily be batch tuned on a local machine. Compared with GBLUP, LightGBM, SVR, DeepGS and DLGWAS, DNNGP is superior to these existing widely used genomic selection (GS) methods. Moreover, DNNGP can generate robust assessments from diverse datasets, including omics data, and quickly incorporate complex and large datasets into usable models, making it a promising and practical approach for straightforward integration into existing GS platforms.

Key words: deep learning, genomic selection, multi-omics data, prediction method

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INTRODUCTION

A major goal of plant breeding is to increase crop yield per unit area, delivering new varieties in a shorter time; this is necessary to meet the challenges of food security for the world's growing population, which is expected to reach 9.5 billion people by 2050 (He and Li, 2020; Qaim, 2020). However, achieving optimum performance has been a consistent challenge for crop breeders because of issues such as extreme weather events, constantly changing environments, plant diseases,

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insect pests, and scarcity of land resources. Accurate predictions of crop performance could help breeders produce varieties with significantly increased agronomic performance and thus satisfy the expected increase in food demand (Bali and Singla, 2022). Genomic prediction allows breeders to estimate genomics-based breeding values using genome-wide marker sets (Meuwissen et al., 2001; Crossa et al., 2017). Genomic prediction is the first step in genomic selection (GS) for plant breeding programs (Bernardo and Yu 2007; Heffner et al., 2009).

GS was first proposed in animal breeding to overcome the shortcomings of marker-assisted selection by predicting breeding values of untested individuals (Meuwissen et al., 2001). GS includes markers distributed across the genome for genomic prediction, irrespective of their effects or significance (Crossa et al., 2017). Because of the dramatic reduction in genotyping costs, GS has rapidly achieved popularity over the past decade and has led to important genetic gains in many plant and animal breeding programs (Bernardo and Yu 2007; Heffner et al., 2009; Crossa et al., 2017). Traditional methods, such as regression-based techniques, are limited in their ability to analyze high-dimensional data and to capture multivariate and complex relationships between predictor and response variables (Meuwissen et al., 2001). Machine learning (ML) methods are capable of capturing non-linear relationships within genotypes while requiring no prior knowledge of potential genetic models (González-Recio et al., 2014; Ma et al., 2014; Abdollahi-Arpanahi et al., 2020). Random forest, a tree-based ML method, outperformed the generalized Poisson ridge regression model in predicting Fusarium head blight severity in 115 spring wheat lines via GS using 1635 single-nucleotide polymorphisms (SNPs) (Montesinos-López et al., 2021). Montesinos-López et al. (2019) used 2038 genotyping-by-sequencing markers to compare a Bayesian threshold genomic best linear unbiased prediction method with two ML-based methods, support vector machine and multi-layer perceptron. The three methods were used to predict days to heading, days to maturity, and plant height (PH) in seven datasets from four elite yield trial nurseries. Compared with extreme gradient boosting, genomic best linear unbiased prediction (GBLUP), and ridge regression best linear unbiased prediction, a method built with light gradient boosting machine (LightGBM) had the best performance in predicting PH, days to tasseling (DTT), and ear weight (EW) in 8652 maize lines using 32 559 SNPs (Yan et al., 2021). Support vector regression (SVR) is a well-known ML method for classification and regression problems. This method can take different types of data as input because it can be used with a variety of kernel functions. One explanation for the necessity of implementing many different ML methods in GS is that there is no universal best prediction method that can be used under all circumstances (i.e., the "no free lunch" theorem; Wolpert and Macready, 1997).

Deep learning (DL) is a recently developed ML technique based on deep multi-layered hidden neural network architecture in which many hidden layers are constructed during network training (Xie et al., 2016). You et al. (2017) applied convolutional neural networks (CNNs) and recurrent neural networks to predict soybean yield from remotely sensed images. The CNN-recurrent neural network method outperformed methods built with random forest, deep fully connected neural networks, and least absolute shrinkage and selection operator (Khaki et al., 2020). Khaki and Wang (2019) designed a deep neural network method to predict maize yield across 2247 locations from 2008 to 2016 using genotype and environmental data in the yield testing stage.

It is now feasible to obtain multi-omics (e.g., genomics, transcriptomics, metabolomics, and proteomics) data at a large scale to aid genomic prediction. This possibility offers new opportunities to predict complex phenotypic traits using various types of data. Several relevant studies have used omics data to perform genomic prediction for complex traits in humans and plants (Vazquez et al., 2016; Dimitrakopoulos et al., 2017; Bellot et al., 2018; Azodi et al., 2020; Hu et al., 2021). Most of these studies concentrated on incorporating multi-omics data into a predictive model to improve prediction accuracy. These data not only provide new layers for genomic prediction but also provide a link between organismal phenotypes and genomic variation that cannot easily be captured at the genome sequence level (Azodi et al., 2020). Deep learning genomic selection (DeepGS), which is based on deep CNNs (Ma et al., 2018), and deep learning genome-wide association study (DLGWAS), which is based on dual CNN streams (Liu et al., 2019), have been proposed as methods for prediction of phenotypes using genotype matrices as the input. Until now, few ML methods have been applied to omics datasets in a range of plant species. It has been necessary to use multiple plant breeding datasets to assess the performance of genomic prediction methods. In this study, we developed a new DL method, deep neural network genomic prediction (DNNGP). The predictive accuracies of methods based on DNNGP and five popular prediction methods (GBLUP, LightGBM, SVR, DeepGS, and DLGWAS) on wheat, maize, and tomato datasets of different sizes were compared to evaluate the performance of the new method. DNNGP is a promising and practical approach that successfully incorporates multi-omics data to predict agronomic traits of large and diverse datasets in the context of GS.

RESULTS

Design of a DL method for genomic prediction

Incorporating multi-omics data, we developed a genomic prediction method, DNNGP, by designing an algorithm in which stacking of multiple linear and non-linear processing units in a layer-wise fashion enables learning of complex representations at different levels of abstraction. Specifically, DNNGP contains three CNN layers: one batch normalization (BN) layer (included to avoid overfitting) and two dropout layers (Figure 1A). The BN layer is used to address internal covariance shifts within input features. In addition to its multilayered, hierarchical structure, DNNGP uses a callback function to direct early stopping to avoid overfitting. Finally, seeking to accelerate the learning process and improve the convergence rate by alleviating the problem of a vanishing gradient, DNNGP uses a non-saturating and non-linear activation function (rectified linear unit [ReLU]). Thus, DNNGP combines the advantages of early stopping and ReLU. Among the other DL-based genomic prediction methods, neither DeepGS nor DLGWAS contains a BN layer (Figure 1B and 1C). DeepGS used an ReLU but not early stopping (Figure 1B), whereas DLGWAS used early stopping but not an ReLU (Figure 1C).

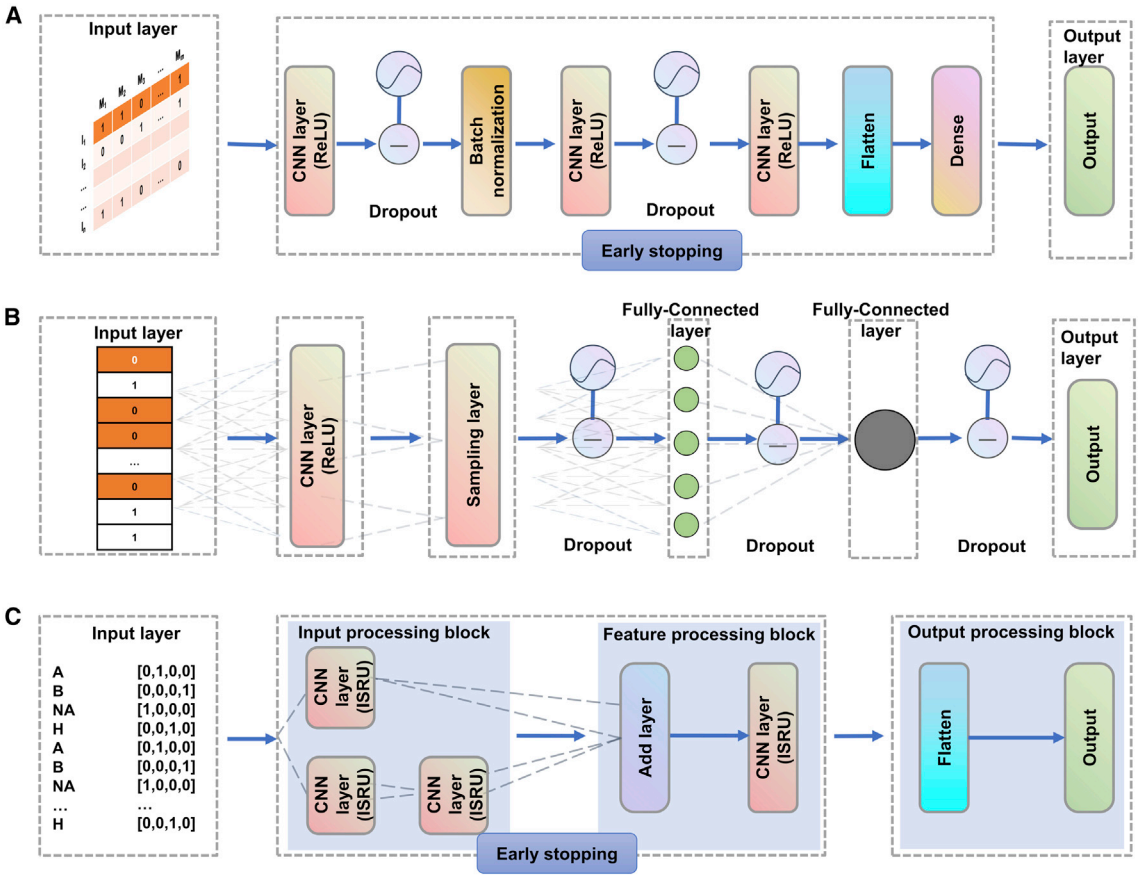

Figure 1. Illustration of the algorithmic frameworks used in DNNGP, DeepGS, and DLGWAS.
(A–C) The algorithmic frameworks used in DNNGP **(A)**, DeepGS **(B)**, and DLGWAS **(C)**. DNNGP, deep neural network for genomic prediction; DeepGS, deep learning genomic selection; and DLGWAS, deep learning genome-wide association study.

Prediction accuracy of DNNGP and comparison with other methods

We compared the prediction accuracy (i.e., the correlation coefficient between predicted and observed values) for GBLUP, LightGBM, SVR, DeepGS, DLGWAS, and DNNGP using four datasets: wheat599, maize1404, wheat2000, and tomato332 (Figure 2 and Supplemental Figures 1 and 2). In addition, 6210 F_1 hybrids of the maize1404 dataset (denoted maize1404_F1) were used to evaluate the performance of different methods (Supplemental Figures 1 and 2). Performance on the wheat599 dataset for grain yield (GY) was evaluated in four environments. The prediction accuracy was evaluated for each method using four, three, six, and one agronomic trait(s) in the maize1404, maize1404_F1, wheat2000, and tomato332 datasets, respectively (methods; Supplemental Table 1). All hyperparameters are shown in Supplemental Table 2.

Differences between methods in prediction accuracy for GY in each environment in the wheat599 dataset may be due to the varying levels of data noise and data structure and, thus, the varying features extracted by different methods. Five of the six methods performed best in the first environment; DLGWAS did not (Figure 2A). DNNGP performed better on average (0.63) across four environments than GBLUP (0.35), LightGBM (0.45), SVR (0.61), DeepGS (0.13), and DLGWAS (0.43). DNNGP outperformed the traditional GBLUP method in all four environments; the prediction accuracies for GBLUP were 64.7%, 65.9%, 164.2%, and 61.5% lower than those of DNNGP in environments 1–4, respectively. When there was no dimensionality reduction (DR) step, GBLUP performance increased from 0.41 to 0.54, 0.39 to 0.51, 0.23 to 0.40, and 0.37 to 0.49 in environments 1–4, respectively; however, GBLUP still had lower performance than DNNGP (Figure 2A; Supplemental Table 3). Compared with DNNGP, LightGBM had 36.3%, 53.2%, 37.2%, and 38.5% lower prediction accuracy in environments 1–4, respectively. DNNGP outperformed SVR in environments 2, 3, and 4 by 14.7%, 1.6%, and 1.5%, respectively; in environment 1, the only exception, SVR outperformed DNNGP by a small margin of 1.4%. DeepGS had the worst prediction accuracy in all four environments. Compared with the other five methods, GBLUP has conceptual differences in the prior assumptions assigned to marker effects. Despite the non-linear modeling structure, DeepGS and DLGWAS performed worse than LightGBM and SVR, indicating that the algorithm architecture of the two DL-based methods was unable to successfully capture the non-linear relationships in the wheat599 dataset. DNNGP had the best performance with respect to the BLUP values (0.62), and DeepGS had the worst (0.04). DNNGP outperformed GBLUP, LightGBM, SVR, DeepGS, and DLGWAS by 78.81%, 60.16%, 3.45%, 1557.64%, and 17.50%, respectively. Overall, these results showed that DNNGP performed better than DL methods on small datasets. The improvement in prediction

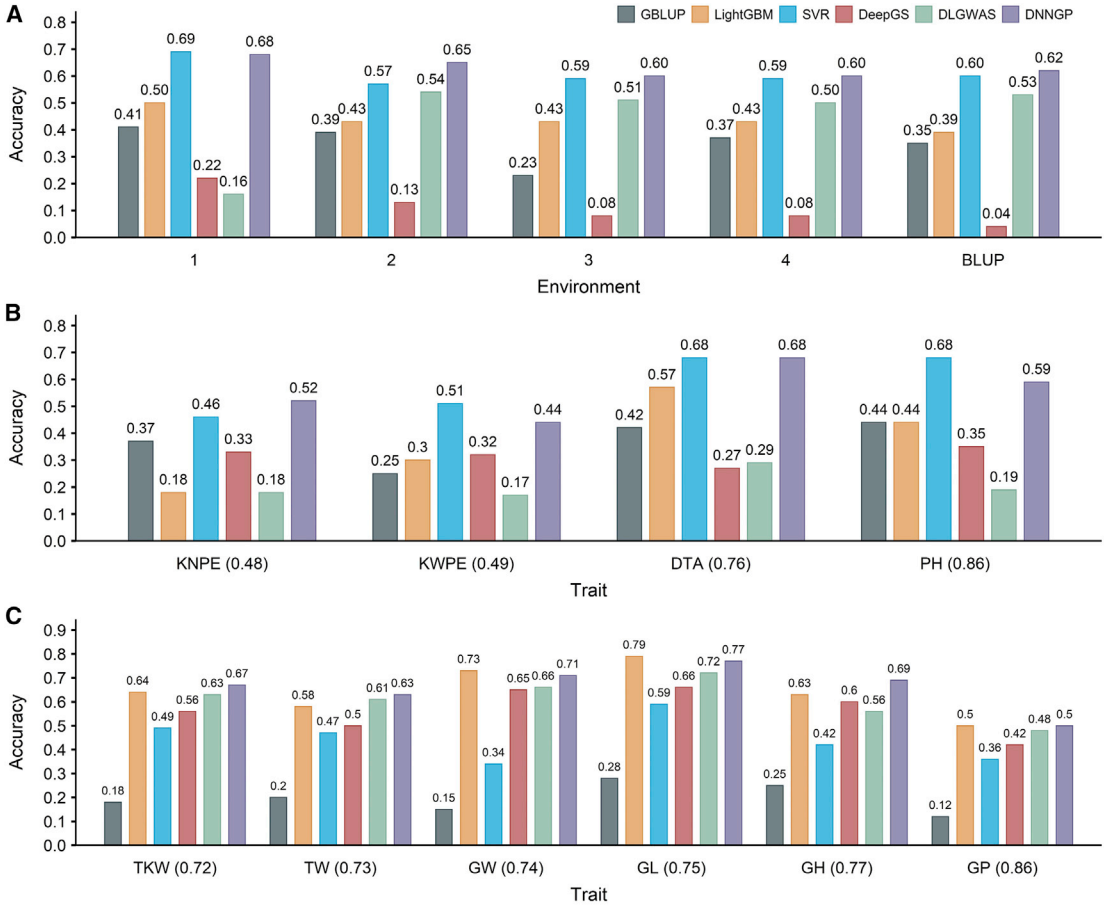

Figure 2. Prediction accuracy of methods built using six different models on three datasets.

(A–C) Prediction accuracy of methods built using six different models on three datasets: wheat599 **(A)**, maize1404 **(B)**, and wheat2000 **(C)**. KNPE, kernel number per ear; KWPE, kernel weight per ear; DTA, date to anther; PH, plant height; TKW, thousand kernel weight; TW, test weight; GW, grain width; GL, grain length; GH, grain hardness; GP, grain protein; GBLUP, genomic best linear unbiased prediction; LightGBM, light gradient boosting machine; SVR, support vector regression; DeepGS, deep learning genomic selection; DLGWAS, deep learning genome-wide association analysis; and DNNGP, deep neural network for genomic prediction. Values in parentheses on the x axis labels are the broad-sense heritability values for the corresponding traits.

accuracy of the DNNGP method was partially due to the multilayered and hierarchical structure that includes three CNN layers, one BN layer, two dropout layers, and ReLU (Figure 1; Supplemental Table 4).

We then determined the prediction accuracies of GBLUP, LightGBM, SVR, DeepGS, DLGWAS, and DNNGP on four phenotypic traits in the maize1404 dataset: kernel number per ear (KNPE), kernel weight per ear (KWPE), date to anther (DTA), and PH (Figure 2B). The highest prediction accuracies were achieved for DTA and PH across all methods, in part because of the high heritability of DTA and PH compared with the other traits (Supplemental Figure 1); thus, the appropriate features were easier to select for DTA and PH than for the other traits. Based on the average prediction accuracy across traits, DLGWAS was the worst method (0.21), and SVR was the best (0.58). DNNGP was the second-best method (0.56), with 4.50% lower accuracy than SVR. DNNGP outperformed GBLUP, LightGBM, DeepGS, and DLGWAS by 50.9%, 48.6%, 75.1%, and 167.0%, respectively. The performance of DNNGP was comparable with that of SVR with respect to DTA predictions but was 12.94% higher for KNPE. In the maize1404_F1 dataset, prediction accuracy was tested on three phenotypic traits: DTT, PH, and EW (Supplemental Figure 2). The highest prediction performance (0.80) was achieved for DTT across all methods. Based on average prediction accuracy across traits, SVR had the worst performance (0.34), DNNGP had the best (0.878), and GBLUP performed slightly worse than DNNGP (0.877). DNNGP outperformed GBLUP, LightGBM, SVR, DeepGS, and DLGWAS by 0.15%, 8.97%, 156.32%, 2.21%, and 12.80%, respectively.

Using the comparatively large dataset wheat2000, we next compared the prediction accuracies of the six methods on six phenotypic traits: thousand kernel weight (TKW), test weight (TW), grain width (GW), grain length (GL), grain hardness (GH), and grain protein (GP) (Figure 2C). GL was predicted with the highest accuracy across all methods. On average, DNNGP outperformed the other five methods to various extents in predicting all traits. These results demonstrated that DNNGP was the best predictive method; it outperformed GBLUP, LightGBM, SVR, DeepGS, and DLGWAS by 234.2%, 2.5%, 48.9%, 16.8%, and

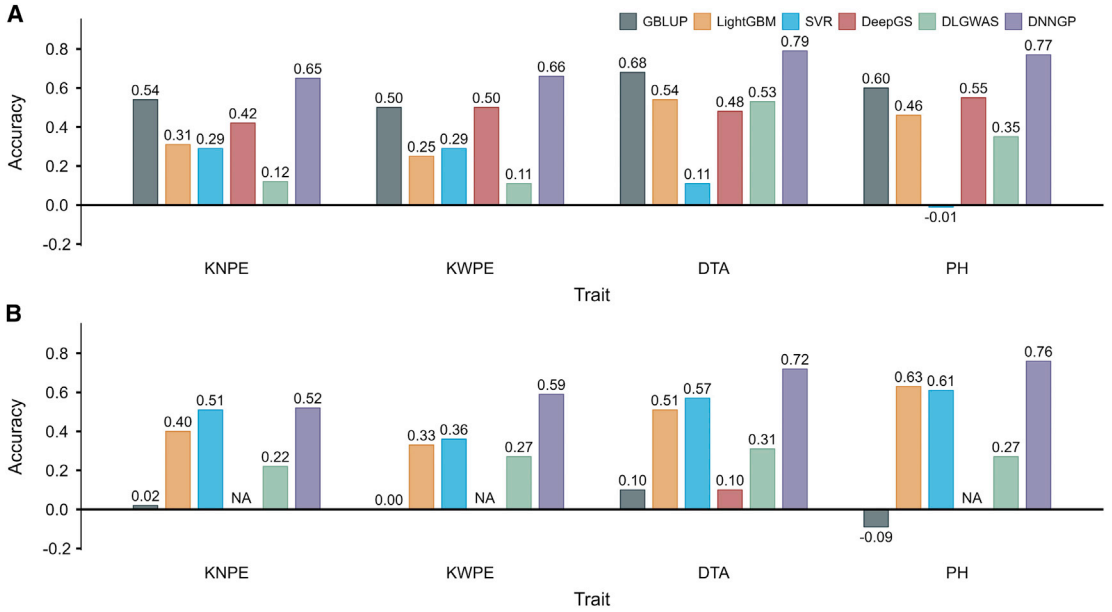

Figure 3. Prediction accuracy of methods built using six different models on maize1404 SNP data and RNA sequencing (RNA-seq) expression data.

(A and B) Prediction accuracy of methods built using six different models on maize1404 SNP data **(A)** and RNA-seq expression data **(B)**. KNPE, kernel number per ear; KWPE, kernel weight per ear; DTA, date to anther; PH, plant height; GBLUP, genomic best linear unbiased prediction; LightGBM, light gradient boosting machine; SVR, support vector regression; DeepGS, deep learning genomic selection; DLGWAS, deep learning genome-wide association analysis; and DNNGP, deep neural network for genomic prediction.

8.2%, respectively. Prediction accuracy in the maize1404, maize1404_F1, and wheat2000 datasets showed that DNNGP gradually outperformed the other predictive methods, extending the performance boost as dataset size increased. By contrast, SVR gradually lagged behind the other methods in performance. The results also suggested that DNNGP had relatively stable performance on all three datasets. The algorithms were also tested with and without DR, a preprocessing step that is commonly used in high-dimensional data analysis. The six methods considered in this study were fairly robust with or without DR, with the exception of GBLUP (Figure 2 and Supplemental Figure 2; Supplemental Table 3). When cross-validation (CV) was performed by calculating a single correlation coefficient across all individuals after all folds were predicted (Supplemental Table 5), the prediction accuracy was comparable with the results presented in Figure 2 and Supplemental Figure 2.

Prediction accuracy using genome-wide omics data

The six methods were compared in terms of their performance for predicting KNPE, KWPE, DTA, and PH using genotypic and transcriptomics data on the maize1404 dataset (Figure 3). In general, DNNGP had the highest prediction accuracy in all datasets and for all traits. For DTA, DNNGP had 619.84% and 16.42% higher prediction accuracy using gene expression and SNP data, respectively, than GBLUP. When genotypic data were used for predictions, the average prediction accuracy of GBLUP, LightGBM, SVR, DeepGS, DLGWAS, and DNNGP was 0.58, 0.39, 0.17, 0.49, 0.28, and 0.72, respectively. Prediction accuracy was generally higher for DTA and PH than for KNPE and KWPE across methods. For example, using the LightGBM method, the average prediction accuracy for DTA and PH using NPs and gene expression data was 0.53 and 0.55, respectively, whereas the average prediction accuracy for KNPE and KWPE was 0.35 and 0.29, respectively. These results may be due to the correlation between heritability and prediction accuracy; KNPE and KWPE showed lower heritability than DTA and PH, so the prediction accuracy was correspondingly lower. Methods built with transcriptomics data generally did not perform worse than those built with SNP data. The gene expression data had many fewer dimensions than the SNP data, but the results indicated that they could still reflect the potential for phenotypic variation.

The tomato332 dataset was analyzed using the six methods to test their performance using diverse omics data (Figure 4). DeepGS failed to generate results on this dataset. DNNGP showed superior performance compared with the other methods across all types of omics data. GBLUP prediction accuracies ranged from 0.08–0.35 for gene expression (GE), structural variation (SV), and insertion or deletion (InDel) data. Using SNP data, the prediction accuracy of LightGBM was 0.13. The SVR and DLGWAS methods built on SNP, GE, and SV data had negative prediction accuracy values, in part because they were unable to adequately capture the relationships between the features and the trait, fruit soluble solid content (SSC), being tested in this dataset.

Non-linear relationships captured by DNNGP in all four datasets

DL enables the generation of promising models to address a broad array of challenges. However, to provide a robust estimate of model accuracy, it is necessary to extend model evaluation beyond the standard analytical framework. Interpretability of DL

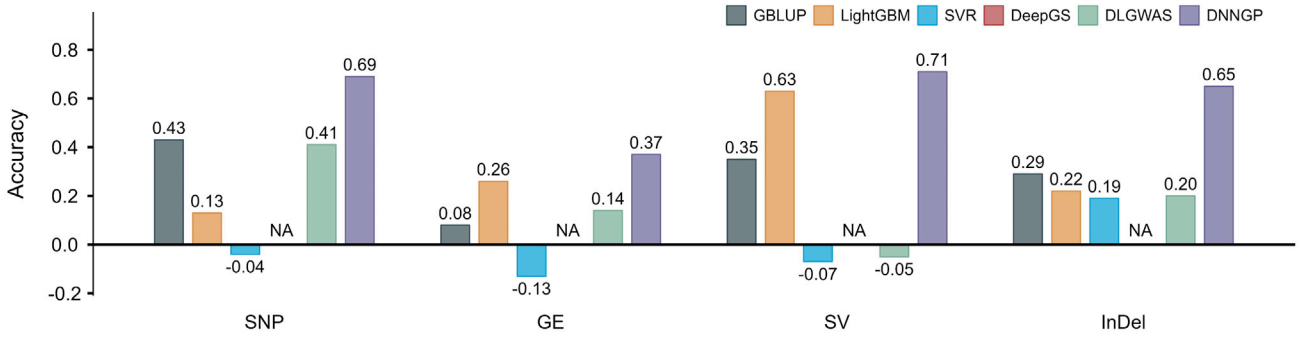

Figure 4. Prediction accuracy of six methods for fruit soluble solid content (SCC) on the tomato332 dataset.
Shown is the prediction accuracy of six methods for fruit soluble solid content (SCC) on the tomato332 dataset. SNP, single-nucleotide polymorphism; GE, gene expression; SV, structural variants; and InDel, insertion or deletion; GBLUP, genomic best linear unbiased prediction; LightGBM, light gradient boosting machine; SVR, support vector regression; DeepGS, deep learning genomic selection; DLGWAS, deep learning genome-wide association analysis; and DNNGP, deep neural network for genomic prediction.

methods is an area of active research in the field of DL. Density plots were used to visualize the relationship between observed values and various features (Figure 5 and Supplemental Figures 3–11; Newman and Furbank, 2021). The red regression lines show the orthogonal distance regression (ODR) fitted from contour lines. The differences in the slopes of regression lines of best fit reflect the genotype–phenotype relationships and demonstrate the different non-linear relationships hidden in the dataset. For the wheat599 dataset, the angle between the fitted regression line and y = x was smallest in environment 1 and largest in environment 3 (Figure 5). In all environments, the density plots reflect the non-linear relationships captured by DNNGP to some extent; this relationship can be succinctly expressed by fitting the regression line. The results illustrate that DNNGP was more likely to generate high-precision trait predictions in heavily populated areas of the phenotypic distribution.

Sample size effects on prediction methods

Population size directly affects the performance of prediction methods and is therefore a major factor to be considered in ML and other classic GS approaches (Xavier et al., 2016; Cericola et al., 2017). We therefore examined the effect of sample size on the prediction performance of methods built with GBLUP, LightGBM, SVR, DeepGS, DLGWAS, and DNNGP. The predicted versus observed values were compared for all six traits in the wheat2000 dataset for all six methods. Ten-fold CV was performed using 100, 200, 500, 800, 1500, 1800, and 2000 randomly selected individuals with a DR step (Table 1) and without a DR step (Supplemental Table 6). Across all methods, increased sample size was associated with improved prediction accuracy. When the sample size increased from 100 to 2000, the prediction accuracy for TKW, TW, GW, GL, GH, and GP improved by 54.0%, 48.0%, 47.7%, 126.2%, 177.3%, and 12.0%, respectively. GBLUP and DNNGP showed gradual improvements in prediction accuracy as the sample size increased. On average, across six traits, when the sample size was 100, DNNGP outperformed GBLUP, LightGBM, SVR, and DLGWAS by 397.86%, 23.24%, 81.39%, and 101.25%, respectively. DeepGS failed to generate predictions when the sample size was very small (∼100–200). This indicated that DNNGP could perform well on small datasets whereas DeepGS could not. When 2000 individuals were used, DNNGP outperformed GBLUP, LightGBM, SVR, DeepGS, and DLGWAS by 234.2%, 2.5%, 48.9%, 16.8%, and 8.2%, respectively (Table 1). DNNGP showed very high prediction accuracy for all six traits. The prediction accuracy of DLGWAS increased with sample size, and the prediction accuracies for TW, GW, and GH were negative when the sample size was ∼100–200 (Table 1 and Supplemental Table 6). This indicates that DLGWAS could not successfully capture the non-linear relationship between genotype and phenotype when the sample size was very small. LightGBM and SVR had relatively stable and consistent prediction accuracy regardless of sample size. When the sample size was ∼1000, GBLUP had a prediction accuracy comparable with that of the LightGBM and SVR methods. However, as sample size increased, GBLUP predictions did not improve significantly. When sample sizes were large, the LightGBM and SVR methods were less robust than GBLUP and had lower prediction accuracies than DNNGP. Predictions were more accurate for GW, GL, and GH than for TKW and TW. These findings suggest that a large number of individuals should be used in GS prediction methods where possible and that DL methods should be implemented to increase prediction accuracy.

SNP number effects on prediction methods

Various platforms have been developed to generate genotypic marker sets in a range of sizes. SNP number reportedly has a significant influence on the performance of GS prediction methods (Heffner et al., 2011). Here we used the wheat2000 dataset to study the effects of SNPs with different p value thresholds on the prediction performance of GBLUP, LightGBM, SVR, DeepGS, DLGWAS, and DNNGP. For each of the tested traits, 10-fold CV was performed using SNPs selected at thresholds of p = 0.1, 0.01, 0.001, 0.0001, or 1e−05 (Supplemental Figure 12). In general, we observed no obvious changes in GS method performance when different thresholds were used. As the p value screening became more stringent, the prediction accuracy of methods built with LightGBM, SVR, DeepGS, DLGWAS, and DNNGP did not change significantly, whereas the performance of GBLUP increased dramatically when the SNP number was reduced. GBLUP is more sensitive than the other five methods. The performance of these methods also suggested that fewer trait-associated SNPs could have the same power as unfiltered

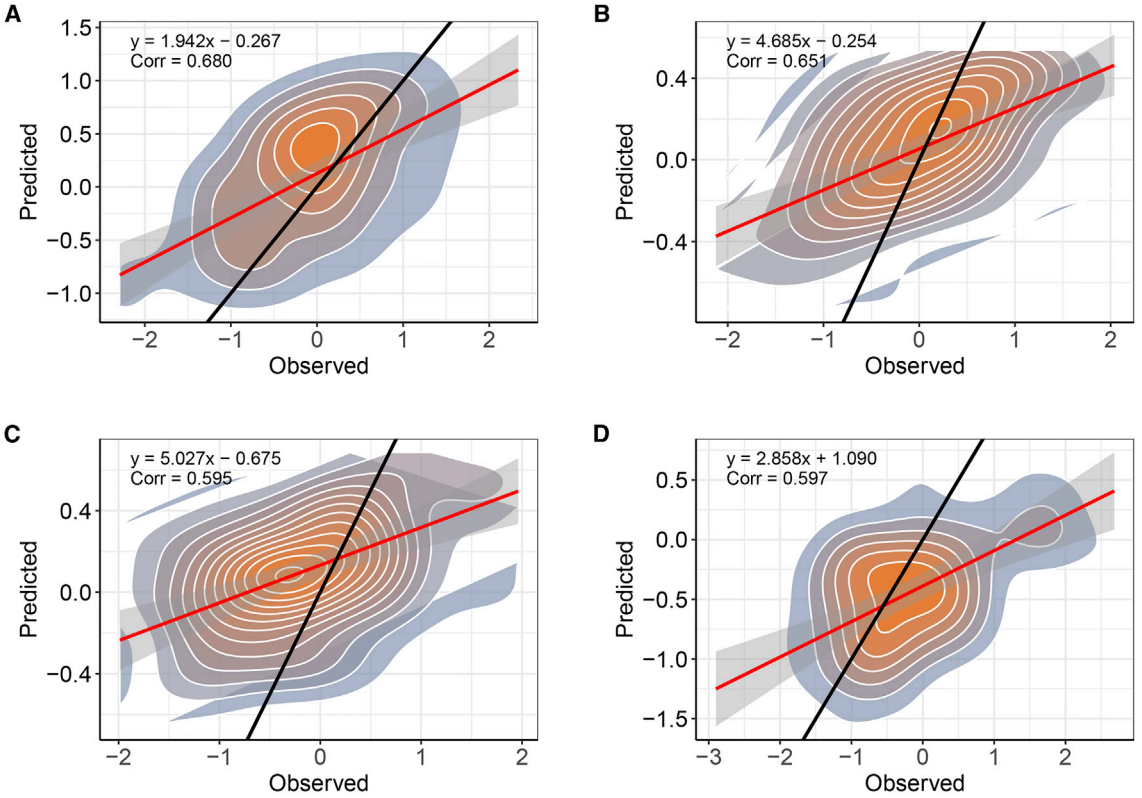

Figure 5. Density plots showing prediction accuracy of DNNGP for the performance of wheat cultivars on the wheat599 dataset in four environments.

(A–D) Density plots showing prediction accuracy of DNNGP for the performance of wheat cultivars on the wheat599 dataset under CV scheme 1 in four environments: 1 **(A)**, 2 **(B)**, 3 **(C)**, and 4 **(D)** (a correlation coefficient was calculated for each fold, and coefficients were then averaged over 10 folds and recorded for a run). Contour lines indicate kernel density. Black lines show the function y = x. Red lines indicate the orthogonal distance regression (ODR) fitted from contour lines; these lines demonstrate that DNNGP captures different non-linear relationships in data from different environments. Negative coordinate values are due to standardization of observed values.

genome-wide SNPs for complex trait prediction. Predictions were generally more accurate for GW, GL, and GH than for TKW and KW, indicating again that traits with higher heritability could be predicted with greater accuracy.

Efficiency of computation time for prediction methods

We next compared the computation time (in seconds) required by each method to generate predictions for four datasets to which DR techniques had been applied (Figure 6). All four datasets could be used with all six methods, with the exception of DeepGS on the tomato332 dataset and maize1404 transcriptome data. The dimension of a dataset is the product of its feature number and the population size, two major factors that determine the computation time required for model building. It is widely assumed that DL methods require longer computation times to complete analysis than traditional methods. In our study, however, DNNGP performed at a level of runtime similar to GBLUP, LightGBM, SVR, and DLGWAS and could be ∼10 times faster than DeepGS. DNNGP took slightly longer on small datasets (i.e., <500 instances) than other methods; however, as the size of dataset used in GS is continuously increasing as a result of advances in sequencing, the outstanding performance of DNNGP on large datasets indicates that it will be a competitive method in the foreseeable future.

DISCUSSION

As an important part of plant breeding, GS facilitates selection of elite breeding lines and accelerates the breeding cycle. There is an urgent need to develop GS methods with increased predictive power. Montesinos-López et al. (2018) found that DL methods were superior to GBLUP in predicting six of nine traits evaluated in wheat and maize. Liu et al. (2019) also demonstrated the superiority of single and dual CNN models over ridge regression best linear unbiased prediction in predicting soybean (*Glycine max* L.) yield, protein content, oil content, moisture content, and height. Similarly, Zingaretti et al. (2020) showed that DL models performed better than conventional linear statistical models in predicting traits with epistatic variances in allopolyploid strawberry species (*Fragaria* × *ananassa*) and blueberry (*Cyanococcus* spp.). These prior studies suggest that there is great potential to increase predictive power for complex traits using DL approaches in plant breeding.

In this study, we developed a DL-based technique, DNNGP, that is capable of uncovering the complex relationships that may be hidden in diverse multi-omics data to predict quantitative traits in plants. In the algorithm architecture of DNNGP, there are three CNN layers, one BN layer, and two dropout layers (Figure 1). Neither DeepGS nor DLGWAS contains a BN layer, which we

Trait	Sample size	GBLUP	LightGBM	SVR	DeepGS	DLGWAS	DNNGP
TKW	100	−0.006	0.531	0.351	N/A	0.333	0.512
	200	0.037	0.209	−0.093	N/A	0.369	0.517
	500	0.202	0.652	0.236	0.415	0.394	0.632
	800	0.189	0.643	0.071	0.390	0.486	0.645
	1000	0.238	0.500	0.323	0.454	0.558	0.646
	1500	0.212	0.630	0.375	0.574	0.566	0.644
	1800	0.256	0.635	0.530	0.410	0.590	0.653
	2000	0.184	0.641	0.489	0.561	0.634	0.670
TW	100	0.237	0.450	0.344	N/A	0.177	0.477
	200	0.248	0.245	0.060	N/A	0.188	0.490
	500	0.181	0.386	0.286	0.219	0.493	0.531
	800	0.171	0.540	0.089	0.507	0.517	0.545
	1000	0.150	0.435	0.369	0.531	0.532	0.544
	1500	0.167	0.540	0.368	0.598	0.567	0.536
	1800	0.175	0.582	0.524	0.594	0.611	0.615
	2000	0.202	0.580	0.469	0.497	0.615	0.630
GW	100	−0.125	0.599	0.378	N/A	0.412	0.564
	200	0.253	0.459	−0.016	N/A	0.416	0.568
	500	0.215	0.515	0.175	0.486	0.453	0.594
	800	0.296	0.557	0.204	0.401	0.493	0.607
	1000	0.208	0.599	0.318	0.560	0.558	0.608
	1500	0.266	0.647	0.371	0.508	0.568	0.697
	1800	0.282	0.676	0.473	0.629	0.617	0.706
	2000	0.154	0.730	0.337	0.648	0.662	0.710
GL	100	0.163	0.565	−0.192	N/A	0.237	0.627
	200	0.164	0.835	0.266	N/A	0.517	0.649
	500	0.301	0.660	0.228	0.639	0.559	0.656
	800	0.318	0.750	0.164	0.578	0.599	0.670
	1000	0.319	0.643	0.494	0.469	0.605	0.670
	1500	0.305	0.638	0.513	0.575	0.630	0.669
	1800	0.323	0.660	0.607	0.480	0.649	0.694
	2000	0.278	0.785	0.586	0.665	0.718	0.766
GH	100	0.110	0.038	0.079	N/A	0.165	0.553
	200	0.252	0.372	0.119	N/A	0.233	0.576
	500	0.221	0.440	0.152	0.609	0.356	0.591
	800	0.244	0.606	0.195	0.632	0.383	0.610
	1000	0.204	0.503	0.356	0.591	0.491	0.612
	1500	0.287	0.481	0.382	0.427	0.505	0.587
	1800	0.291	0.629	0.502	0.521	0.556	0.668
	2000	0.250	0.631	0.419	0.601	0.557	0.688
GP	100	0.234	0.294	0.723	N/A	0.192	0.320
	200	−0.006	0.289	0.482	N/A	0.283	0.324
	500	0.094	0.322	0.170	0.518	0.295	0.334

Table 1. Prediction accuracy of methods built using six different models and subsets of instances from the wheat2000 dataset.

(Continued on next page)

Trait	Sample size	GBLUP	LightGBM	SVR	DeepGS	DLGWAS	DNNGP
	800	0.042	0.448	0.205	0.636	0.363	0.479
	1000	0.176	0.490	0.237	0.634	0.497	0.520
	1500	0.153	0.553	0.342	0.576	0.509	0.523
	1800	0.122	0.457	0.416	0.444	0.433	0.441
	2000	0.118	0.499	0.360	0.420	0.476	0.498

Table 1. *Continued*

TKW, thousand kernel weight; TW, test weight; GW, grain width; GL, grain length; GH, grain hardness; GP, grain protein; GBLUP, genomic best linear unbiased prediction; LightGBM, light gradient boosting machine; SVR, support vector regression; DeepGS, deep learning genomic selection; DLGWAS, deep learning genome-wide association analysis; DNNGP, deep neural network for genomic prediction.

included to prevent overfitting. Removing the BN layer from DNNGP led to a dramatic reduction in prediction accuracy; there was no convergent solution in the training set in most cases, meaning that no final prediction was available when the model was implemented without the BN layer (Supplemental Table 4). The multilayered hierarchical structure of DNNGP allows dynamic feature learning from the raw data, giving DNNGP the ability to extract low-, mid-, and high-level features. An internal covariance shift refers to a change in the distribution of the values of hidden units, which slows down convergence by forcing the learning rate toward a small value; this requires careful parameter initialization. Previous studies have established that BN can be used to address internal covariance shifts among features (Khan et al., 2020). In addition to the multilayered hierarchical structure, DNNGP was designed to avoid overfitting by employing early stopping (a callback function) (Figure 1). To improve the convergence rate (specifically through alleviating the problem of a vanishing gradient and to accelerate the learning process), DNNGP implements a non-saturating and non-linear activation function, ReLU. The superior performance of DNNGP can probably be largely attributed to its use of multiple feature extraction stages that can automatically learn representations from data.

We investigated applications of DNNGP and compared the results with those of other GS methods for predicting complex traits in four different datasets. The algorithm architecture design of the DNNGP method gives it flexibility in how it can be applied; there are few existing methods that utilize multi-omics data in ML or DL to improve prediction power. DNNGP can handle complex inputs such as SNP, transcriptomics, and proteomics data. By contrast, GBLUP cannot use transcriptomics or proteomics data directly. To use GBLUP with such data as the input, we built a kinship matrix separately for the transcriptomics data but found that the performance was poor (Figures 3 and 4). Using the maize1404 transcriptomics data or the four examined inputs (i.e., SNPs, GE data, SVs, and InDelsindels) of the tomato332 dataset, the DeepGS model did not yield coherent results (Figures 3 and 4). We tested several strategies to correct this problem, including adjusting the phenotypes (by scaling and centralizing) or hyperparameters (batch size and learning rate), but DeepGS did not provide any phenotype predictions; this was consistent with its designed purpose, predicting phenotypes only from genotypic rather than multi-omics data (Ma et al., 2018).

Our new predictive method performed as well as or better than several commonly used methods, i.e., a linear model (GBLUP), two ML-based methods (LightGBM and SVR), and two DL-based methods (DeepGS and DLGWAS), in a broad range of tasks. The high prediction accuracy of DNNGP compared with GBLUP in CV can be attributed to the presence of hidden layers that automatically capture complex hidden interactions without prior specification. It is therefore understandable that, in all cases, DNNGP performed as well or even better than the commonly used linear model (GBLUP). Hyperparameter optimization in multilayer DL models strongly affects performance (Khan et al., 2020), but such optimization has high computational costs. For ML methods (LightGBM and SVR), optimization can be effectively performed using a cross-validated grid search (Young et al., 2015). The DNNGP method was designed with three CNN layers, which reduces computational stress. Each trait requires various combinations of hyperparameters (Supplemental Table 2). However, unlike DLGWAS and DeepGS, the hyperparameters can easily be batch tuned using a random search on a local machine (Bergstra and Bengio, 2012), minimizing the computational cost. In general, the DL-based methods performed worse than the ML-based methods and DNNGP in predicting agronomic traits; this was especially true when the training set was small (Figure 2A; Table 1, and Supplemental Table 6). Because of the algorithm design of DNNGP (Figure 1), its performance was greatly improved as the amount of data increased, and its computation time was greatly reduced, particularly on large datasets (Figures 2, 3, 4, and 6; Table 1 and Supplemental Table 6).

Increasing the statistical power of prediction methods is an area of active research that aims to enable more efficient allocation of breeding resources and facilitate optimal design of GS experiments in breeding programs. To this end, we evaluated the prediction accuracy of all six methods on the wheat2000 dataset using a selection of SNPs (Supplemental Figure 12) and subsets of instances (Table 1 and Supplemental Table 6). This experiment was designed to establish the optimal number of individuals in the training population and number of markers in the feature set that are necessary to maximize model accuracy. The results showed that the most important component was the training sample size; DNNGP predictions continuously improved with increasing training dataset size. However, DNNGP also had outstanding prediction accuracy when sample sizes were small (Figure 2; Table 1 and Supplemental Table 6). Our results and some prior studies demonstrate that DL-based methods can achieve equivalent or superior accuracy in GS compared with traditional linear models, even with a relatively small training set. Overfitting is a concern when using DL with a small training set; this results from failure of the model to learn general patterns in the data.

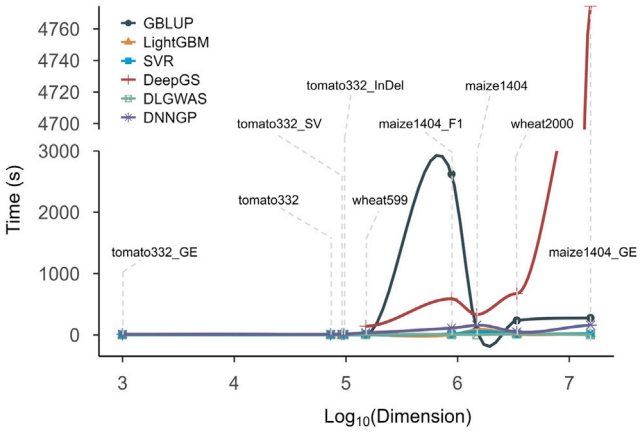

Figure 6. Computation time required by six methods to generate predictions for four datasets using different omics features.

Dimension was calculated as the sample size multiplied by the number of features. GE, gene expression data from RNA-seq; SV, structural variants; InDel, insertion/deletion mutation; GBLUP, genomic best linear unbiased prediction; LightGBM, light gradient boosting machine; SVR, support vector regression; DeepGS, deep learning genomic selection; DLGWAS, deep learning genome-wide association analysis; and DNNGP, deep neural network for genomic prediction.

To avoid overfitting, in DNNGP, we used a BN layer, dropout layer, and early stopping, which involves removal of a fixed number of neurons during model training.

Plant scientists are often interested in understanding the biological meaning underlying predictive methods. This is difficult in DL-based methods because of the "black box" nature of neural networks and the large number of layers and neurons involved in model training. We mapped the non-linear relationships of several traits using density plots (Figure 5 and Supplemental Figures 3–11). This approach plays a vital role in understanding DL methods. However, running and interpreting DL-based methods require a background in computer science and statistics, and they may therefore require additional expertise or collaborations. DL approaches are promising for use in the context of GS-aided plant breeding programs. This study opens new avenues for the application of DL to prediction of complex traits in plant breeding. In subsequent studies, more effort can be focused on feature extraction when processing large omics datasets; it may also be useful to implement a graph neural network to mine the internal relationships among features (Zhu et al., 2022). If successful, such approaches could then replace the DR step currently used by DNNGP and would enable a huge number of features to be directly processed within the algorithm. The improved performance of DNNGP for genomic prediction using multi-omics datasets, combined with its ease of use and high computational efficiency, indicate that DL-based approaches can be widely adopted in genomic prediction studies.

There is now active demand for DL-based approaches that can be incorporated into the plant breeding toolkit for use in large-scale breeding programs to improve genetic gains for quantitative traits (Xu et al., 2022; Yan and Wang, 2022). Here, we introduced a new method, DNNGP, to predict quantitative traits from multi-omics data in the context of GS in plants. We evaluated the performance of DNNGP in comparison with five popular methods (GBLUP, LightGBM, SVR, DeepGS, and DLGWAS), using four different datasets of differing sizes. DNNGP showed several advantages: (1) it can handle complex input such as genomics, transcriptomics, and proteomics data; (2) it has a BN layer in its algorithm architecture that helps avoid overfitting and ensures that there are convergent solutions from the training sets while implementing early stopping and ReLU to further avoid overfitting and improve the convergence rate; (3) it performed as well as or better than a commonly used linear model (GBLUP), ML-based methods (LightGBM and SVR), and DL-based methods (DLGWAS and DeepGS) in a broad range of prediction tasks; (4) it had a similar runtime to GBLUP, LightGBM, SVR, and DLGWAS and performed ~10 times faster than DeepGS; and (5) unlike the aforementioned DL methods, the model hyperparameters of DNNGP can be easily tuned in batch based on configuration on a local machine. The application of DL to plant breeding is an active reseach area, and insights gained through development, testing, and application of DNNGP in this study would benefit future wide applications of DL in crop breeding.

METHODS

Datasets used for genomic prediction

Four datasets, representing various population sizes and species with different reproductive systems, were used in this study: wheat599, maize1404, wheat2000, and tomato332. The first dataset, wheat599, is a collection of 599 historical wheat lines from the International Maize and Wheat Improvement Center (CIMMYT) Global Wheat Program (McLaren et al., 2005). This program previously conducted numerous international trials in a wide range of wheat-producing environments. The environments represented in these trials were divided into four main agroclimatic regions (also called mega-environments) that were precisely defined and widely used by the CIMMYT global breeding program. The dataset comprised phenotypic, genotypic, and pedigree information (Crossa et al., 2010) and has been used previously to evaluate several predictive methods (de Los Campos et al., 2009; Crossa et al., 2010, 2014; Pérez-Rodríguez et al., 2010; Caamal-Pat et al., 2021). Each wheat line was genotyped using 1447 diversity array technology (DArT) markers generated by Triticarte (Canberra, ACT, Australia; https://www.diversityarrays.com). The DArT markers were recorded as present (1) or absent (0) for each line. Markers with a minor allele frequency of less than 0.05 were removed, and missing genotypes were imputed with samples from the marginal distribution of marker genotypes as follows:

$$x_{ij} \sim Bernoulli(\hat{p}_j)$$

where $\hat{p}_j$ is the estimated allele frequency computed from the non-missing genotypes. After quality control, 1279 markers were retained. DR is a preprocessing step that is used in DL-based approaches to remove redundant features and noisy or irrelevant data, improving feature accuracy and reducing training time. A principal-component analysis (PCA)-based approach was used here to extract the top 95% of feature variance (Burges, 2009). PCA was used for several reasons: (1) it is a common DR approach, (2) it is appropriate for the high-dimensional input data types DNNGP was designed to process, and (3) it has been shown to substantially reduce DL computation time. After DR, 251 principal components (PCs) were retained for analysis. The phenotypic trait considered for this dataset was the average GY of the 599 wheat lines evaluated in each of the four mega-environments. To exclude environmental noise in the phenotypic data, the BLUP values were calculated by the mixed linear model in the R package "lme4" and were also used to evaluate the methods (Supplemental Figure 1).

The second dataset, maize1404, consisted of 1404 progeny descended from 24 Chinese elite inbred maize lines using the complete-diallel plus unbalanced breeding-derived inter-cross design (Liu et al., 2020). The 1404 lines and 30 checks, together with all founder parents, were evaluated in 5 locations for 23 agronomic traits. The BLUP values for each line were used to reduce environmental noise in the phenotypic data. Each line was re-sequenced with 11-fold coverage to obtain 11.8 million SNP markers. After quality control for missing data, heterozygosity, and minor allele frequency, 6 730 418 SNPs were retained. PCA was used for DR, and 1056 PCs were retained for use in predictions. Four agronomic traits were considered: DTA, which is related to flowering time; PH, which is related to plant architecture; and KNPE and KWPE, which are related to yield (Supplemental Figure 1). We also used published RNA sequencing data from 392 randomly selected individuals in the maize1404 dataset (Liu et al., 2020), which included expression level values for 39 484 genes. PCA was used for DR, and 209 PCs were retained. In addition to inbred lines, 6210 F_1 hybrids of maize1404 (207 maternal × 30 paternal lines, denoted maize1404_F_1) were inferred by combining the maternal and paternal alleles derived from the complete-diallel plus unbalanced breeding-derived inter-cross population (Yan et al., 2021). Three phenotypic traits (DTT, PH, and EW) were measured in 5 locations. To reduce the influence of the environment, BLUP values were calculated for the phenotypic data in each F_1 hybrid and each parental line in the 5 locations over 2 years using the mixed linear model in the R package "lme4"; these BLUP values were then used for subsequent analysis (Supplemental Figure 1). There were 32 599 haplotypic tag SNPs evenly distributed in the genome. After DR with PCA, 142 PCs were retained.

The third dataset, wheat2000, contained 2000 Iranian bread wheat (*Triticum aestivum*) landraces from the CIMMYT wheat gene bank; these were genotyped with 33 709 DArT markers (Crossa et al., 2016). Each allele was recorded as 1 (present) or 0 (absent) in each accession. After DR with PCA, 1691 PCs were retained for analysis. Six agronomic traits were evaluated: TKW, TW, GL, GW, GH, and GP (Supplemental Figure 1). To evaluate the influence of marker number on prediction accuracy for all six methods, subsets of 33 709 DArT markers were selected using different p values at thresholds of 0.1, 0.01, 0.001, 0.0001, and 1e−05 (Student's t-test). To evaluate the influence of sample number on prediction accuracy, 100, 200, 500, 800, 1000, 1500, and 1800 samples were randomly selected from the wheat2000 dataset.

The fourth dataset, tomato332, was a call set (designated TGG1.1–332) from the tomato graph pangenome consisting of 6 971 059 SNPs, 657 549 InDels, 51 155 GEs, and 54 838 SVs (Zhou et al., 2022). After DR with PCA, we retained 220, 289, 3, and 277 PCs from SNPs, InDels, GEs, and SVs, respectively. SSC, an important yield and flavor trait (Duangjit et al., 2016), was predicted in further analyses (Supplemental Figure 1).

DNNGP algorithm

DNNGP is based on a deep CNN; the architecture includes one input layer, three convolutional layers, one BN layer, two dropout layers, one flattening layer, one dense layer, and one output layer (Figure 1A). The input layer receives m markers (or other types of omics data) for n individuals as an $n \times m$ matrix. The three CNN layers are the core building blocks in which the majority of computation occurs. The output of a convolutional layer can be defined as an integral transformation and is represented as follows (Sandhu et al., 2021):

$$s(t) = (f * k)(t) = \sum_x k(t - x) f(x)$$

where k represents the kernel, convolution is the transformation of f into $s(t)$, and the operation is performed over an infinite number of copies f shifting over the kernel.

There are four basic hyperparameters that affect CNN capabilities and must be set before training: (1) number of filters, (2) kernel size, (3) activation function, and (4) regularization. The number of filters affects the depth of the output (e.g., three distinct filters would yield three different feature maps, creating a depth of three). In this study, we set 64 filters in each convolution layer. Kernel size indicates the length of the convolution window, controls the step for the input sequence being read each time, and corresponds to the feature map through the convolution process. The kernel size here was set to 4 in each convolution layer. Activation functions convert the sum of input signals into output signals; different activation functions have different mechanisms. At present, commonly used activation functions are sigmoid, ReLU, linear, and softmax. After each convolution operation, DNNGP applies an ReLU transformation to the feature map, introducing non-linearity to the model. The ReLU directly yields the value when the input is greater than 0 and yields a value of 0 when the input is less than or equal to 0. This function can be represented with the following formula:

$$h(x) = \begin{Bmatrix} x \ (x > 0) \\ 0 \ (x \leq 0) \end{Bmatrix}$$

Regularization is a common solution to overfitting in ML and can reduce model complexity. The L2 regularization is applied in DNNGP to modify the learning algorithm, reducing generalization error (but not training error). It is expressed as follows:

$$J = J_0 + \alpha \sum_w w^2$$

where J_0 represents the initial loss function, α represents the regularization coefficient, and w represents the square root of the square sum of each element in the weight vector ω, which follows the normal distribution with mean zero. DNNGP also includes a BN layer, which can prevent overfitting and control gradient explosion to prevent the gradient from disappearing. The input units of two dropout layers were randomly set to 0 at each step during training to prevent overfitting. Data from the third CNN can be flattened by the flattening layer and then classified by the dense layer. The output layer is responsible for testing set predictions. DNNGP also includes a callback function called early stopping, which fine-tunes the learning rate to optimize the model (Khan et al., 2020).

Methods used for comparison

The predictive performance of DNNGP was compared with the performance of five popular methods: GBLUP, LightGBM, SVR, DeepGS, and DLGWAS. GBLUP is an efficient method for predicting genotypic values of target populations using genome-wide markers. It replaces the kinship matrix based on pedigree relationships used in traditional BLUP by constructing a genomic relationship matrix that is estimated by genome-wide markers. This approach more accurately predicts breeding values. The genomic relationship matrix is calculated as follows:

$$\boldsymbol{G} = \frac{\boldsymbol{Z}\boldsymbol{Z}'}{2\sum_{i=1}^{m} p_i(1 - p_i)}$$

where $\boldsymbol{Z}$ is the centralized genotype matrix, which can be calculated using the marker genotypic matrix $\boldsymbol{M}$ (VanRaden, 2008). The genotype matrix $\boldsymbol{M}$ is an $n \times m$ dimensional matrix in which n is the sample size, m is the number of markers, and the element in $\boldsymbol{M}$ is −1 (homozygote), 0 (heterozygote), or 1 (other homozygote). p_i is the frequency of the minor allele of the ith marker. The matrix is defined as

$$\boldsymbol{P} = 2(p_i - 0.5)1_{(m \times 1)}, \ i = 1, 2, \ldots, m,$$

and $\boldsymbol{Z} = \boldsymbol{M} - \boldsymbol{P}$. When maize1404 GE and tomato332 GE, SV, and InDel data were used, kinship was estimated using the correlations between individuals calculated from the corresponding variants (Wu et al., 2022).

After obtaining the kinship matrix, the effects of individuals can be determined using a mixed-model equation (Henderson, 1953). GBLUP was implemented in the "BGLR" R package (Pérez-Rodríguez and de Los Campos, 2014).

LightGBM was developed by Microsoft in 2017 and is among the most popular ML methods for big data analysis. It has ultra-high efficiency in handling structured data with high-dimensional features and large sample sizes (Ke et al., 2017). LightGBM is more powerful than extreme gradient boosting, with a short training time and lower memory usage. LightGBM is mainly characterized by a decision tree algorithm using gradient-based one-side sampling, exclusive feature bundling (EFB), and a histogram and leaf-wise growth strategy with a depth limit. Gradient-based one-side sampling trains the single gradient on a specific feature of the sample as the weight of the sample. The specific process is as follows:

1. Select a percentage (**a**) of samples with larger gradients as the training samples for instances with larger gradient values.
2. From the samples (1−**a**) with small gradient values, randomly select a percentage (**b**) as the training samples for instances with small gradient values.
3. For samples with a small gradient (i.e., **b**(1−**a**) samples), multiply by (1−**a**)/**b** when calculating the information gain.

EFB can be used to solve the problem of data sparseness by effectively combining several features. The steps for performing EFB are as follows:

1. Construct a weighted, undirected graph with the vertices as features and the edges as weights; the weights are related to the conflict between two features.
2. Sort the vertices in descending order based on the degree of each node. A greater degree indicates greater conflict with other features.
3. Traverse each feature and assign it to an existing feature set or create a new feature set to minimize the overall conflict.

SVR is a technique for solving regression problems based on the principle of support vector machine (Maenhout et al., 2007). Linear SVR can be described as

$$\min_{\omega,\, b} \frac{1}{2}\|\boldsymbol{\omega}\|^2 + C\sum_{i=1}^{n} l_\varepsilon(f(\boldsymbol{x}_i) - y_i)$$

where $\boldsymbol{\omega}$ is the weight vector, b is the intercept, C is the regularization constant, n is the sample size, a vector $\boldsymbol{x}_i$ and a y_i value form each training sample i as a couple, and l_ε is the ε-insensitive loss function. Introducing the relaxation variables ξ_i and $\widehat{\xi}_i$ yields the following formula:

$$\min_{\omega,b} \frac{1}{2}\|\boldsymbol{\omega}\|^2 + C\sum_{i=1}^{n}(\xi_i + \widehat{\xi}_i)$$

$$s.t.\ \boldsymbol{Y} - (\boldsymbol{A\omega} + b\boldsymbol{e}) \le \varepsilon\boldsymbol{e} + \xi, \xi \ge 0, (\boldsymbol{A\omega} + b\boldsymbol{e}) - \boldsymbol{Y} \le \varepsilon\boldsymbol{e} + \xi^*, \xi^* \ge 0$$

A Lagrange function is then introduced to obtain the dual problem:

$$\max_{\alpha,\beta} -\frac{1}{2}(\alpha - \beta)\boldsymbol{AA}^T(\alpha - \beta) - \varepsilon\boldsymbol{e}^T(\alpha + \beta) + \boldsymbol{Y}^T(\alpha - \beta),$$

$$s.t.\ \boldsymbol{e}^T(\alpha - \beta) = 0,\ 0 \le \alpha, \beta \le C$$

where $\boldsymbol{\omega}$ is the weight vector, C is the regularization constant, b is the intercept, n is the sample size, $\boldsymbol{Y}$ is the output data matrix, $\boldsymbol{A}$ is the input data matrix, $\boldsymbol{e}$ is a vector with n-dimensional data, ε is the relaxation factor, ξ and ξ^* are relaxation variables, and α and $\beta \in R^n$ are Lagrange multipliers. By solving the dual problem, the final regression function $f(x) = \boldsymbol{\omega}^T x + b$ is obtained. LightGBM and SVR methods were implemented in Python version 3.6.5 using the "lightgbm" and "Scikit-learn" (2019) modules, respectively, and run on a GPU-equipped Linux workstation.

DeepGS is a DL method built with CNN layers (Ma et al., 2018). It includes one input layer, one CNN layer, one sampling layer, three dropout layers, two fully connected layers, and one output layer (Figure 1B). The ReLU function is applied in the convolutional and first fully connected layers. The loss function is the mean absolute error index:

$$MAE = \frac{1}{n}\sum_{k=1}^{n}|predict_k - obs_k|$$

where n denotes the number of individuals in the training dataset, and $predict_k$ and obs_k represent the predicted and observed phenotypic values, respectively, of the kth individual.

DLGWAS contains one input layer, one input-processing block, one feature-processing block, and one output-processing block (Figure 1C; Liu et al., 2019). The input-processing block has a dual CNN stream. The first stacked CNN stream contains two feedforward CNN layers with kernel sizes of 4 and 20. The second single CNN stream contains one CNN layer with a kernel size of 4. The feature-processing block with one CNN layer was added to integrate all outputs and process genotype features in the block. The output-processing block contains a flattening layer and a single neuron to generate the final predicted phenotypes. The inverse square root unit activation function was applied to add constraints to the predicted phenotype values and increase the learning rate.

A 10-fold CV scheme was used to evaluate the performance of the six methods on the four datasets (Resende et al., 2012; Crossa et al., 2016; Gianola and Schön, 2016; Qiu et al., 2016). Each dataset was randomly split into 10 equally sized folds, and then each fold (comprising 10% of the population) in turn had phenotypic values removed and was used as the testing set. Prediction accuracy was defined as the Pearson's correlation coefficient calculated for each fold/testing set and then averaged over 10 folds to be recorded for that run. The statistical properties of this strategy were extensively evaluated by Bengio and Grandvalet (2004). This strategy, denoted CV scheme 1 to facilitate illustration, has been adopted in widely available Python modules such as "Scikit-learn" (2019) and is extremely common in ML research (Ballard et al., 2020; Mbatchou et al., 2021). We tested another method of 10-fold CV (denoted CV scheme 2): calculating a single correlation coefficient across all individuals after all folds are predicted (Supplemental Table 5). This maintains a consistent sample size for all calculations. Both strategies of CV were repeated for 100 runs for all prediction methods. The prediction accuracy was the averaged correlation coefficient over 100 runs. The results from the CV scheme 1 analysis are presented in the main tables and figures and in Supplemental Figures 3–6. The results from the CV scheme 2 analysis are presented in Supplemental Table 5 and Supplemental Figures 7–11). We present results obtained using the full version of DNNGP (in the main figures and tables) as well as results obtained from a test iteration of DNNGP in which no DR step was implemented (Supplemental Tables 3 and 6). To ensure that methods were compared using the exact same datasets (with the same dimensions), the results from GBLUP, LightGBM, SVR, DeepGS, and DLGWAS with a DR step are presented in the main figures and tables, and those with no DR step are presented in Supplemental Tables 3 and 6.

To visualize the relationship of mean changes between observed values and various features (Newman and Furbank 2021), density plots were constructed for the predicted and observed values in the testing sets. We applied the "odregress" function from the R package "pracma" to

build the ODR. This is a regression technique in which observational errors on dependent and independent variables are taken into account. The R package "ggplot2" was used to visualize the density plot. The contour lines indicate kernel density, black lines show the function y = x, and red lines show the fitted lines from ODR. We present the density plots of DNNGP using all four datasets in Figure 5 and Supplemental Figures 3–6 for CV scheme 1 and in Supplemental Figures 7–11 for CV scheme 2.

Heritability estimation

The broad-sense heritability of each trait was calculated as follows:

$$H^2 = \frac{\sigma_G^2}{\sigma_G^2 + \sigma_\varepsilon^2}$$

where σ_G^2 is the genotypic variance and σ_ε^2 is the residual variance. The variances were estimated by the mixed linear model in the R package "lme4."

CODE AND DATA AVAILABILITY

The DNNGP scripts are available in the release package on github (https://github.com/AIBreeding/DNNGP/releases/download/v1.0.0/DNNGP-v1.0.0.zip). The datasets used in this study are available from https://github.com/AIBreeding/DNNGP/blob/main/example-data.tgz. Hyperparameters are available from https://github.com/AIBreeding/DNNGP/tree/main/hyperparameters. The detailed user manual is available from https://github.com/AIBreeding/DNNGP/blob/main/DNNGP-usermanual.pdf.

SUPPLEMENTAL INFORMATION

Supplemental information is available at *Molecular Plant Online*.

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AUTHOR CONTRIBUTIONS

H.L. conceived and designed the study. K.W. and H.L. analyzed the data and wrote the manuscript. M.A.A., A.R., J.C., and S.H. edited the manuscript. All authors read and approved the final manuscript.

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REFERENCES

Abdollahi-Arpanahi, R., Gianola, D., and Peñagaricano, F. (2020). Deep learning versus parametric and ensemble methods for genomic prediction of complex phenotypes. Genet. Sel. Evol. **52**:1–15.

Azodi, C.B., Pardo, J., VanBuren, R., de Los Campos, G., and Shiu, S.H. (2020). Transcriptome-based prediction of complex traits in maize. Plant Cell **32**:139–151.

Bali, N., and Singla, A. (2022). Emerging trends in machine learning to predict crop yield and study its influential factors: a survey. Arch. Comput. Methods Eng. **29**:95–112.

Ballard, Z.S., Joung, H.A., Goncharov, A., Liang, J., Nugroho, K., Di Carlo, D., Garner, O.B., and Ozcan, A. (2020). Deep learning-enabled point-of-care sensing using multiplexed paper-based sensors. NPJ Digit. Med. **3**:66.

Bellot, P., de Los Campos, G., and Pérez-Enciso, M. (2018). Can deep learning improve genomic prediction of complex human traits? Genetics **210**:809–819.

Bengio, Y., and Grandvalet, Y. (2004). No unbiased estimator of the variance of k-fold cross-validation. J. Mach. Learn. Res. **5**:1089–1105.

Bergstra, J., and Bengio, Y. (2012). Random search for hyper-parameter optimization. J. Mach. Learn. Res. **13**:281–305.

Bernardo, R., and Yu, J. (2007). Prospects for genomewide selection for quantitative traits in maize. Crop Sci. **47**:1082–1090.

Burges, C.J.C. (2009). Dimension reduction: a guided tour. FNT. in Machine Learning **2**:275–364.

Caamal-Pat, D., Pérez-Rodríguez, P., Crossa, J., Velasco-Cruz, C., Pérez-Elizalde, S., and Vázquez-Peña, M. (2021). lme4GS: an R-package for genomic selection. Front. Genet. **12**:982.

Cericola, F., Jahoor, A., Orabi, J., Andersen, J.R., Janss, L.L., and Jensen, J. (2017). Optimizing training population size and genotyping strategy for genomic prediction using association study results and pedigree information. A case of study in advanced wheat breeding lines. PLoS One **12**:e0169606.

Crossa, J., Campos, G.d.L., Pérez, P., Gianola, D., Burgueño, J., Araus, J.L., Makumbi, D., Singh, R.P., Dreisigacker, S., Yan, J., et al. (2010). Prediction of genetic values of quantitative traits in plant breeding using pedigree and molecular markers. Genetics **186**:713–724.

Crossa, J., Jarquín, D., Franco, J., Pérez-Rodríguez, P., Burgueño, J., Saint-Pierre, C., Vikram, P., Sansaloni, C., Petroli, C., Akdemir, D., et al. (2016). Genomic prediction of gene bank wheat landraces. G3-Genes Genom Genet **6**:1819–1834.

Crossa, J., Pérez-Rodríguez, P., Cuevas, J., Montesinos-López, O., Jarquín, D., de los Campos, G., Burgueño, J., González-Camacho, J.M., Pérez-Elizalde, S., Beyene, Y., et al. (2017). Genomic selection in plant breeding: methods, models, and perspectives. Trends Plant Sci. **22**:961–975.

Crossa, J., Pérez, P., Hickey, J., Burgueño, J., Ornella, L., Cerón-Rojas, J., Zhang, X., Dreisigacker, S., Babu, R., Li, Y., et al. (2014). Genomic prediction in CIMMYT maize and wheat breeding programs. Heredity **112**:48–60.

de Los Campos, G., Naya, H., Gianola, D., Crossa, J., Legarra, A., Manfredi, E., Weigel, K., and Cotes, J.M. (2009). Predicting quantitative traits with regression models for dense molecular markers and pedigree. Genetics **182**:375–385.

Dimitrakopoulos, L., Prassas, I., Diamandis, E.P., and Charames, G.S. (2017). Onco-proteogenomics: multi-omics level data integration for accurate phenotype prediction. Crit. Rev. Clin. Lab Sci. **54**:414–432.

Duangjit, J., Causse, M., and Sauvage, C. (2016). Efficiency of genomic selection for tomato fruit quality. Mol. Breeding **36**:1–16.

Gianola, D., and Schön, C.C. (2016). Cross-validation without doing cross-validation in genome-enabled prediction. G3: Genes, Genomes, Genetics **6**:3107–3128.

González-Recio, O., Rosa, G.J., and Gianola, D. (2014). Machine learning methods and predictive ability metrics for genome-wide prediction of complex traits. Livest. Sci. **166**:217–231.

He, T., and Li, C. (2020). Harness the power of genomic selection and the potential of germplasm in crop breeding for global food security in the era with rapid climate change. The Crop Journal **8**:688–700.

Heffner, E.L., Jannink, J.L., and Sorrells, M.E. (2011). Genomic selection accuracy using multifamily prediction models in a wheat breeding program. The Plant Genome Journal **4**:65–75.

Heffner, E.L., Sorrells, M.E., and Jannink, J.L. (2009). Genomic selection for crop improvement. Crop Sci. **49**:1–12.

Henderson, C.R. (1953). Estimation of variance and covariance components. Biometrics **9**:226–252.

Hu, H., Campbell, M.T., Yeats, T.H., Zheng, X., Runcie, D.E., Covarrubias-Pazaran, G., Broeckling, C., Yao, L., Caffe-Treml, M., Gutiérrez, L.A., et al. (2021). Multi-omics prediction of oat agronomic and seed nutritional traits across environments and in distantly related populations. Theor. Appl. Genet. **134**:4043–4054.

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., and Liu, T. (2017). LightGBM: a highly efficient gradient boosting decision tree. Adv. Neural Inf. Process. Syst. **30**:3149–3157.

Khaki, S., and Wang, L. (2019). Crop yield prediction using deep neural networks. Front. Plant Sci. **10**:621.

Khaki, S., Wang, L., and Archontoulis, S.V. (2019). A CNN-RNN framework for crop yield prediction. Front. Plant Sci. **10**:1750.

Khan, A., Sohail, A., Zahoora, U., and Qureshi, A.S. (2020). A survey of the recent architectures of deep convolutional neural networks. Artif. Intell. Rev. **53**:5455–5516.

Liu, H.J., Wang, X., Xiao, Y., Luo, J., Qiao, F., Yang, W., Zhang, R., Meng, Y., Sun, J., Yan, S., et al. (2020). CUBIC: an atlas of genetic architecture promises directed maize improvement. Genome Biol. **21**:1–17.

Liu, Y., Wang, D., He, F., Wang, J., Joshi, T., and Xu, D. (2019). Phenotype prediction and genome-wide association study using deep convolutional neural network of soybean. Front. Genet. **10**:1091.

Ma, C., Xin, M., Feldmann, K.A., and Wang, X. (2014). Machine learning-based differential network analysis: a study of stress-responsive transcriptomes in Arabidopsis. Plant Cell **26**:520–537.

Ma, W., Qiu, Z., Song, J., Li, J., Cheng, Q., Zhai, J., and Ma, C. (2018). A deep convolutional neural network approach for predicting phenotypes from genotypes. Planta **248**:1307–1318.

Maenhout, S., De Baets, B., Haesaert, G., and Van Bockstaele, E. (2007). Support vector machine regression for the prediction of maize hybrid performance. Theor. Appl. Genet. **115**:1003–1013.

Mbatchou, J., Barnard, L., Backman, J., Marcketta, A., Kosmicki, J.A., Ziyatdinov, A., Benner, C., O'Dushlaine, C., Barber, M., Boutkov, B., et al. (2021). Computationally efficient whole-genome regression for quantitative and binary traits. Nat. Genet. **53**:1097–1103.

McLaren, C.G., Bruskiewich, R.M., Portugal, A.M., and Cosico, A.B. (2005). The International Rice Information System. A platform for meta-analysis of rice crop data. Plant Physiol. **139**:637–642.

Meuwissen, T.H., Hayes, B.J., and Goddard, M.E. (2001). Prediction of total genetic value using genome-wide dense marker maps. Genetics **157**:1819–1829.

Montesinos-López, O.A., Martín-Vallejo, J., Crossa, J., Gianola, D., Hernández-Suárez, C.M., Montesinos-López, A., Juliana, P., and Singh, R. (2019). A benchmarking between deep learning, support vector machine and Bayesian threshold best linear unbiased prediction for predicting ordinal traits in plant breeding. G3: Genes, Genomes, Genetics **9**:601–618.

Montesinos-López, O.A., Montesinos-López, A., Crossa, J., Gianola, D., Hernández-Suárez, C.M., and Martín-Vallejo, J. (2018). Multi-trait, multi-environment deep learning modeling for genomic-enabled prediction of plant traits. G3: Genes, Genomes, Genetics **8**:3829–3840.

Montesinos-López, O.A., Montesinos-López, J.C., Salazar, E., Barron, J.A., Montesinos-Lopez, A., Buenrostro-Mariscal, R., and Crossa, J. (2021). Application of a Poisson deep neural network model for the prediction of count data in genome-based prediction. Plant Genome **14**:e20118.

Newman, S.J., and Furbank, R.T. (2021). Explainable machine learning models of major crop traits from satellite-monitored continent-wide field trial data. Nat. Plants **7**:1354–1363.

Pérez, P., and de Los Campos, G. (2014). Genome-wide regression and prediction with the BGLR statistical package. Genetics **198**:483–495.

Pérez, P., de Los Campos, G., Crossa, J., and Gianola, D. (2010). Genomic-enabled prediction based on molecular markers and pedigree using the Bayesian linear regression package in R. Plant Genome **3**:106–116.

Qaim, M. (2020). Role of new plant breeding technologies for food security and sustainable agricultural development. Appl. Econ. Perspect. Policy **42**:129–150.

Qiu, Z., Cheng, Q., Song, J., Tang, Y., and Ma, C. (2016). Application of machine learning-based classification to genomic selection and performance improvement. In International Conference on Intelligent Computing, *9771* (Springer, Cham), pp. 412–421.

Resende, M.F.R., Muñoz, P., Resende, M.D.V., Garrick, D.J., Fernando, R.L., Davis, J.M., Jokela, E.J., Martin, T.A., Peter, G.F., and Kirst, M. (2012). Accuracy of genomic selection methods in a standard data set of loblolly pine (*Pinus taeda* L.). Genetics **190**:1503–1510.

Sandhu, K.S., Lozada, D.N., Zhang, Z., Pumphrey, M.O., and Carter, A.H. (2021). Deep learning for predicting complex traits in spring wheat breeding program. Front. Plant Sci. **11**:613325.

Scikit-learn. (2019). Cross-validation: evaluating estimator performance. https://scikit-learn.org/stable/modules/cross_validation.html. [Accessed 14 May 2019]. accessed on.

VanRaden, P.M. (2008). Efficient methods to compute genomic predictions. J. Dairy Sci. **91**:4414–4423.

Vazquez, A.I., Veturi, Y., Behring, M., Shrestha, S., Kirst, M., Resende, M.F.R., Jr., and de Los Campos, G. (2016). Increased proportion of variance explained and prediction accuracy of survival of breast cancer patients with use of whole-genome multiomic profiles. Genetics **203**:1425–1438.

Wolpert, D.H., and Macready, W.G. (1997). No free lunch theorems for optimization. IEEE Trans. Evol. Comput. **1**:67–82.

Wu, P.Y., Stich, B., Weisweiler, M., Shrestha, A., Erban, A., Westhoff, P., and Inghelandt, D.V. (2022). Improvement of prediction ability by integrating multi-omic datasets in barley. BMC Genom. **23**:200–215.

Xavier, A., Muir, W.M., and Rainey, K.M. (2016). Assessing predictive properties of genome-wide selection in soybeans. G3: Genes, Genomes, Genetics **6**:2611–2616.

Xie, J., Girshick, R., and Farhadi, A. (2016). Unsupervised deep embedding for clustering analysis International conference on machine learning. PMLR **48**:478–487.

Xu, Y., Zhang, X., Li, H., Zheng, H., Zhang, J., Olsen, M.S., Varshney, R.K., Prasanna, B.M., and Qian, Q. (2022). Smart breeding driven by big data, artificial intelligence and integrated genomic-enviromic prediction. Mol. Plant **15**:1664–1695. https://doi.org/10.1016/j.molp.2022.09.001.

Yan, J., and Wang, X. (2022). Unsupervised and semi-supervised learning: the next frontier in machine learning for plant systems biology. Plant J. **111**:1527–1538.

Yan, J., Xu, Y., Cheng, Q., Jiang, S., Wang, Q., Xiao, Y., Ma, C., Yan, J., and Wang, X. (2021). LightGBM: accelerated genomically designed crop breeding through ensemble learning. Genome Biol. **22**:271.

You, J., Li, X., Low, M., Lobell, D., and Ermon, S. (2017). Deep Gaussian process for crop yield prediction based on remote sensing data. In

Thirty-First AAAI Conference on Artificial Intelligence, *31* (Palo Alto, CA, USA: Stanford University), pp. 4559–4565.

Young, S.R., Rose, D.C., Karnowski, T.P., Lim, S.H., and Patton, R.M. (2015). Optimizing deep learning hyper-parameters through an evolutionary algorithm. In Proceedings of the Workshop on Machine Learning in High-Performance Computing Environments, *4* (NY, USA: Association for Computing Machinery New York), pp. 1–5.

Zhou, Y., Zhang, Z., Bao, Z., Li, H., Lyu, Y., Zan, Y., Wu, Y., Cheng, L., Fang, Y., Wu, K., et al. (2022). Graph pangenome captures missing heritability and empowers tomato breeding. Nature **606**:527–534.

Zhu, J., Wang, J., Han, W., and Xu, D. (2022). Neural relational inference to learn long-range allosteric interactions in proteins from molecular dynamics simulations. Nat. Commun. **13**:1661.

Zingaretti, L.M., Gezan, S.A., Ferrão, L.F.V., Osorio, L.F., Monfort, A., Muñoz, P.R., Whitaker, V.M., and Pérez-Enciso, M. (2020). Exploring deep learning for complex trait genomic prediction in polyploid outcrossing species. Front. Plant Sci. **11**:25.